

Mega-Trading Architecture

Public market data to generative order-flow scenarios, with explicit data lineage and profiling evidence.

(a) DATA PLANE

Public trades

Binance monthly archives
32 liquid USDT symbols
2023-05 to 2026-04

Causal features

action, side, price depth
relative size, interarrival time

(b) TOKENIZATION

Composite event token

one integer ID per event
quantile-fitted bins

Dataset contract

block size 512, stride 256
579 realized token IDs

(c) MODEL CORE

Llama-style decoder

RoPE + RMSNorm
GQA self-attention
SwiGLU feed-forward

Submitted shape

20 layers, hidden 1024
16 Q heads, 4 KV heads

(d) TRAINING

Config-driven runs

Hydra + Accelerate
bf16 mixed precision
Muon + AdamW

Observability

MLflow metrics
checkpoints + manifests

(e) EVALUATION

Chronological backtest

held-out future windows
loss, PPL, top-k

Rollout checks

generated vs. real facts
ablation gates

(f) OUTPUTS

Review artifacts

PDF report
static Space
backtest dashboard

Inference path

top-k rollout
decoded event features

Evidence band

profiling: optimizer CPU window
323.645 ms to 20.018 ms

lineage: tokenizer.json
+ tokens-numpy.json

kernel: Triton GQA forward
156.23 TFLOP/s

Evidence links in the report point to the GitHub docs for data-plane design, training-plane design, kernels, training configs, ablation results, and performance profiling.